

Supplementary Information

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Mechanistic simulation of heritable expression states, cell division and drug response in single-cell populations. Supplementary Notes 1–11, Supplementary Figures 1–3 and Supplementary Tables 1–4.

Supplementary Note 1: Stationary laws used for validation

Birth-death. For $\emptyset \xrightarrow{k} X \xrightarrow{\gamma} \emptyset$ the stationary distribution is Poisson with mean k/γ .

Telegraph model. With activation k_{on} , inactivation k_{off} , transcription k_{tx} from the active state and degradation γ , and $a = k_{\text{on}}/\gamma$, $b = k_{\text{off}}/\gamma$, $\lambda = k_{\text{tx}}/\gamma$, the stationary distribution is the Beta-Poisson mixture

$$P(n) = \frac{\lambda^n}{n!} \frac{\Gamma(a+n)\Gamma(a+b)}{\Gamma(a+b+n)\Gamma(a)} {}_1F_1(a+n; a+b+n; -\lambda),$$

whose mean is $\lambda a/(a+b)$ and whose variance is $\mu + \lambda^2 ab/[(a+b)^2(a+b+1)]$. We evaluate ${}_1F_1(\alpha; \beta; -\lambda)$ through Kummer’s transformation $e^{-\lambda} {}_1F_1(\beta - \alpha; \beta; \lambda)$, whose series has positive terms; partial sums are rescaled by 10^{-200} whenever they exceed 10^{200} so that the log of the sum is accumulated without overflow.

Bursty protein. Protein produced in geometric bursts of mean size b arriving at rate a and degraded at rate γ is negative binomial with shape a/γ and success probability $1/(1+b)$. The package emulates bursts through an mRNA of lifetime $1/(200\gamma)$ translated at rate $200\gamma b$.

Supplementary Note 2: Kernel details and correctness tests

The direct method draws the waiting time from $\text{Exp}(a_0)$ and the channel by linear search over the cumulative propensities; after each event only the propensities of reactions that depend on a changed species (the dependency graph) are recomputed, and the running total is refreshed every 512 events. Fixed-step tau-leaping draws $\text{Poisson}(a_j\tau)$ firings for every channel; the strict variant errors when a species would become negative, the hybrid variant discards the leap and simulates that interval with the direct method. Discarding is conditioned on the realised leap, so the accepted steps of the hybrid variant follow a truncated Poisson law and the kernel is approximate; Fig. 2d measures the resulting discrepancy against the exact stationary laws at two step sizes. Adaptive tau-leaping classifies as critical every channel within $n_c = 10$ firings of exhausting a reactant, selects τ from the non-critical channels through the bounds $\max(\varepsilon x_i/g_i, 1)/|\mu_i|$ and $\max(\varepsilon x_i/g_i, 1)^2/\sigma_i^2$, draws the time to the next critical event from $\text{Exp}(a_0^{\text{crit}})$, fires at most one critical channel, halves τ on rejection, and performs 100 exact events whenever $\tau < 10/a_0$.

Supplementary Table 1 lists the tests of the package test suite that validate the kernels: Kolmogorov-Smirnov distances below 0.015 (direct) or 0.02 (approximate kernels) to the Poisson, Beta-Poisson and negative-binomial

laws with 8 000 to 12 000 cells; agreement of the adaptive kernel with the direct method on a stiff dimerisation network within 5% in the mean; identical results with one and with several threads.

Supplementary Note 3: Parameters of all simulations

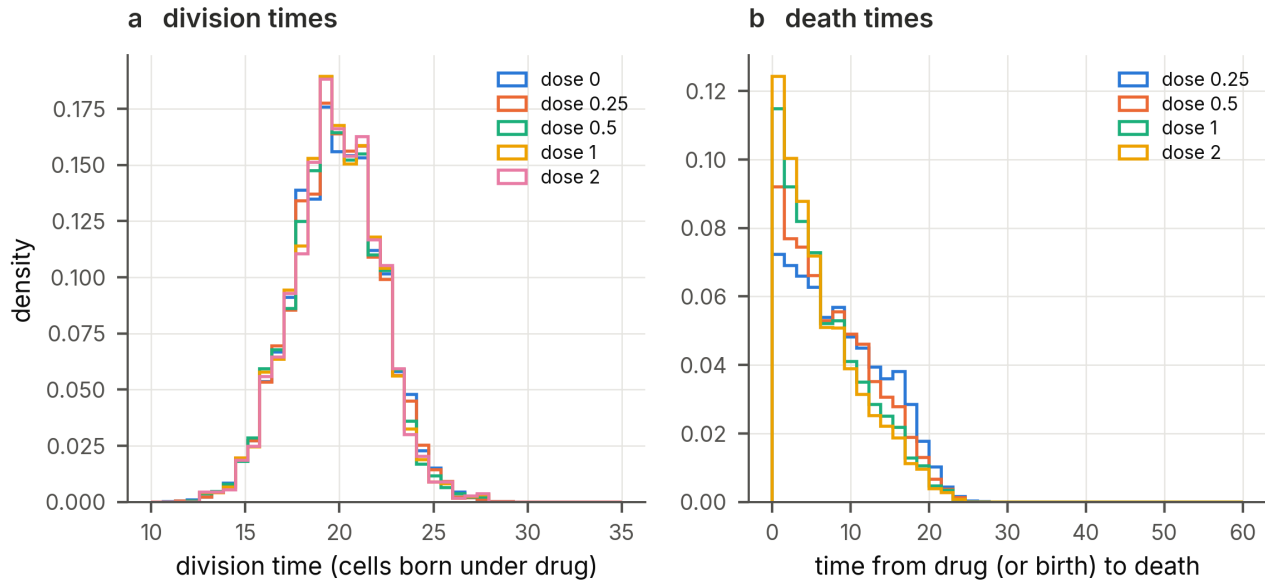
Supplementary Table 2. Parameters of every simulation in this Article.

Figure	Model	Parameters
2a	birth-death	$k = 12, \gamma = 1, T = 15$
2b	telegraph	$(k_{\text{on}}, k_{\text{off}}, k_{\text{tx}}) = (0.4, 0.6, 12)$ and $(0.05, 0.5, 40), \gamma = 1, T = 40$
2c	bursty protein	$a = 1.5, b = 6, \gamma = 1, T = 25$
2d-e	telegraph	$(0.4, 0.6, 12), n = 20\,000$
2f	random telegraph GRNs	$n_{\text{act}} = n_{\text{genes}}, n_{\text{inh}} = n_{\text{genes}}/2, \lambda = \ln 2/20, dt = 0.1, T = 20$
3	telegraph gene with protein	$k_{\text{tx}} = 30, k_{\text{dm}} = 1, k_{\text{tl}} = 4, k_{\text{dp}} = 0.2, k_{\text{on}} = k_{\text{off}}$ as indicated; $\lambda = \ln 2/20$, sizer $V_{\text{div}} = 2$ (cv 0.05), $\sigma_f = 0.02$, replication at 50% of the cycle where indicated
3e	stable protein	$k_{\text{on}} = k_{\text{off}} = 5, k_{\text{tx}} = 20, k_{\text{dm}} = 1, k_{\text{tl}} = 2, k_{\text{dp}} = 0.05$
4	resistance gene	as Fig. 3 with $(k_{\text{on}}, k_{\text{off}}) = (0.002, 0.02)$ (memory, $p_{\text{on}} \approx 9\%$) or $(0.05, 0.5)$ (fast, same p_{on}); founders burnt in for 300 time units at constant size; death $h_{\text{max}} = 0.15, \text{EC}_{50} = 0.5, m = 2, K = 150, q = 4$; drug from $t = 40$ of the treatment stage; drug-induced variant $(0.0002, 0.02)$ with $k_{\text{on}} \rightarrow k_{\text{on}}(1 + 100d)$; fitness cost variant: growth $\times(1 - 0.5 c_P^4 / (150^4 + c_P^4))$; memory disruption: switching $\times 20$ on $[0, 40)$ or $[0, 120)$; schedule scan: 300 founders, 240 time units, 20-time-unit exposures
5a	40 independent telegraph genes	$k_{\text{on}} = k_{\text{off}} \in [10^{-3}, 1]$ (log-spaced), $k_{\text{tx}} = 20, k_{\text{dm}} = 0.5$; 40 founders, 6 doublings

Figure	Model	Parameters
5b-c	random GRN	30 genes, 30 activations, 15 inhibitions, $k_{tx} \in [5, 30]$, $K \in [2, 10]$, $n = 2$, basal 0.05; 1 500 cells; sequencing capture 0.15 (cv 0.3)
5d	random GRN	20 genes, 22 activations, 10 inhibitions; knockdown factor 0.05; 800 cells
6a-b	telegraph	truth (0.3, 0.6, 20); priors log-uniform on $[0.01, 10]$, $[0.01, 10]$, $[1, 200]$; 200 (a) / 120 (b) particles
6c	resistance gene	truth $h_{max} = 0.5$, $K = 150$; priors log-uniform on $[0.05, 5]$ and $[20, 1000]$; 100 particles
7	resistance gene, hours	cell cycle 24 h (sizer, cv 0.1), $k_{tx} = 30$, $k_{dm} = 1$, $k_{tl} = 4$, $k_{dp} = 0.2$ per h, $K = 150$, $q = 4$, growth-arrest Hill coefficient 2; fitted: k_{on} , $k_{off} \in [10^{-3}, 0.3]$, $h_{max} \in [0.005, 0.5]$ per h, $EC_{50} \in [3, 40]$ μ M, $m \in [1, 6]$, $IC_{50} \in [1, 40]$ μ M; 300 founders burnt in for 10 cycles, 48 h drug-free, 72 h drug; 160 Latin-hypercube points and 80 Nelder-Mead iterations with seed 1
8a-d	melanoma-like	$k_{tx} = 3$, $k_{dm} = 0.1$, $k_{tl} = 0.4$, $k_{dp} = 0.02$ per h; net doubling 4 weeks, memory 5 generations, pre-resistant fraction 0.005, $h_{max} = 0.0023$ per h, $EC_{50} = 0.3$, $m = 2$, protection $K = 150$, $q = 4$, growth arrest $IC_{50} = 0.3$ with the same protection, $k_{off} \rightarrow k_{off}/(1 + 9d)$, fitness cost 0.5, partial protection: additional unprotected growth inhibition with $IC_{50} = 1$; 2 000 founders with promoter states drawn from the stationary distribution, 60 weeks, dt 4 h; progression at $1.73 \times$ nadir after the 8-week lead-in

Figure	Model	Parameters
8e-g	MGMT model	same expression kinetics; net doubling 40 days, memory 4 generations, MGMT-expressing fraction 0.01 or 0.30, $h_{\max} = 0.03$ per h at peak, $EC_{50} = 0.4$ of the standard bolus peak, $m = 2$, $K = 150$, $q = 4$; elimination half-life 2.1 h; MGMT consumption $k_{dp} \rightarrow k_{dp}(1 + 20d)$; 1 500 founders at the stationary promoter distribution, six 28-day cycles, dt 1 h

33 Supplementary Note 4: Additional persister simulations



34
35 **Supplementary Fig. 1 | Single-cell timing distributions underlying Fig. 4c.** **a**, Division times of cells
36 born under drug and **b**, times from drug start (or from birth, for cells born under drug) to death, per dose.

37 The division-time distribution is set by the sizer and is the same at every dose, because in this model the drug
38 kills but does not slow growth. Times to death are broad (coefficient of variation 0.72 at dose 0.5 and 0.82 at
39 dose 2) and their mean shifts by much less than the population decay rate, because most deaths occur among
40 low-expressing cells whose hazard is already close to $h_{\max}$ at dose 0.5 ($EC_{50} = 0.5$, $m = 2$); raising the dose
41 mainly shortens the survival of cells with intermediate protection. The dose dependence of population decay is
42 therefore carried by the fraction of cells that are protected, not by the kinetics of death of unprotected cells,
43 which is the interpretation¹ give of their single-cell tracking data.

44 Under continuous dosing at the highest dose (release period 0, dose 2 in Fig. 4f), the long-term growth rate is

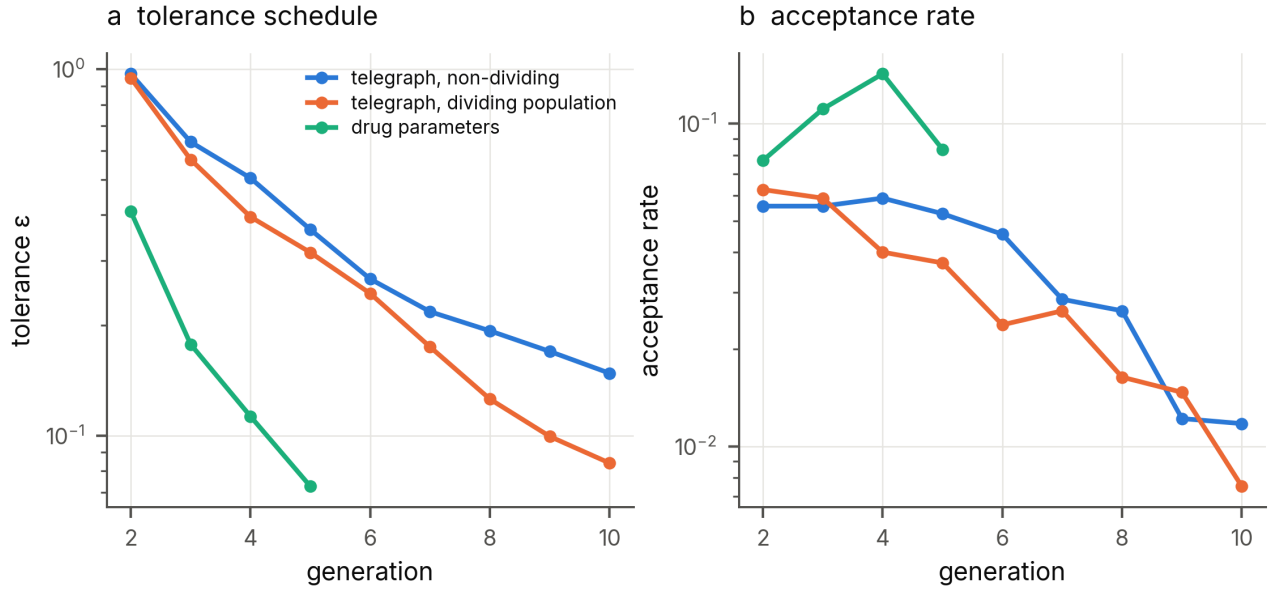
45 +0.004 per time unit for pre-existing tolerance, -0.013 when the resistant state carries a 50% growth cost and
 46 +0.012 for drug-induced tolerance; the schedules with the lowest long-term growth rate are release period 0, dose
 47 2 (growth rate +0.004 per time unit), release period 10, dose 2 (growth rate -0.016 per time unit) and release
 48 period 10, dose 1 (growth rate +0.009 per time unit) respectively. For pre-existing tolerance without a cost, every
 49 release period raises the net growth rate; with the fitness cost, release periods of 5 and 10 time units are within
 50 0.003 per time unit of continuous dosing at doses of 1 and 2; for drug-induced tolerance, dose 1 gives a lower net
 51 growth rate than dose 2 at every release period, and release periods of 20 time units or more raise the growth rate
 52 in every model. All values are single realisations with 300 founder cells.

53 **Supplementary Note 5: Benchmark details**

54 Network inference methods: absolute Pearson and Spearman correlations of $\log(1 + x)$ (Pearson) or ranks
 55 (Spearman); GENIE3 as random-forest importances (200 trees, `sqrt` features) fitted on $\log(1 + x)$ with scikit-learn.
 56 Imputation: kNN-smoothing (one step, $k = 15$, 10 principal components of the Freeman-Tukey transformed,
 57 library-size normalised counts) and MAGIC with default parameters on square-root normalised counts. Metrics:
 58 area under the precision-recall and receiver-operating curves against the undirected (correlation) or directed
 59 (GENIE3) ground-truth adjacency. The cell-cycle-regressed dataset is the residual of $\log(1 + c)$ (concentration c)
 60 after ordinary least-squares regression on gene copy number, cell age and age squared, per gene.

61 **Supplementary Note 6: Inference diagnostics**

62 **Summary statistics and distances.** For Fig. 6a,b the distance between a simulated and the observed sample
 63 of counts is their Wasserstein-1 distance², computed as the mean absolute difference between the two empirical
 64 quantile functions on a grid of 201 probabilities, divided by the observed mean; each simulated sample contains
 65 600 cells (a) or the final snapshot of 500 founders grown for 80 time units (b), against 2 000 observed cells. The
 66 distance between two independent samples at the true parameters (the noise floor) is reported in the summary
 67 tables of the repository. For Fig. 6c the summaries of a treated population are the log surviving fraction on a
 68 grid of 25 time points from the start of the drug to 60 time units later (populations that went extinct contribute
 69 a floor of 10^{-3}) and the concordance of death fates between sisters born in the 20 time units before the drug; the
 70 distance is the root-mean-square difference of these summaries. Each generation's tolerance is the median of the
 71 previous generation's accepted distances, particles are perturbed with a Gaussian kernel of variance twice the
 72 weighted variance of the previous generation, and the first generation samples the prior.



73

74 **Supplementary Fig. 2 | ABC-SMC diagnostics. a**, Tolerance and **b**, acceptance rate per generation for the
75 three runs of Fig. 6.

76 The run on non-dividing cells (10 generations, 200 particles) reached a final tolerance of 0.148 with an acceptance
77 rate of 1% in the last generation; the division-aware run (10 generations, 120 particles, each particle simulating
78 a population of 250 founders for 80 time units) reached 0.0841 at 1%; the drug-parameter run (5 generations,
79 100 particles, 200 founders each) reached 0.0727 at 8%. Posterior predictive checks were made by simulating the
80 forward model at the posterior median (Fig. 6d for the drug parameters); the posterior median rather than the
81 mean is reported throughout because the posteriors are asymmetric on the log scale.

82 Supplementary Note 7: Runtime

83 Supplementary Table 3 lists the wall-clock times of the population simulations of Fig. 2f (200 steps of $dt = 0.1$ for
84 random telegraph networks of 10, 50 and 100 genes, four threads of an Intel Xeon processor at 2.1 GHz, Julia
85 1.10). Throughput in cell-steps per second falls with the number of reactions because every step of every cell
86 has to sample the events of all channels; the hybrid kernel is 2 to 5 times faster than the direct method at these
87 tolerances, the gap widening with the number of reactions. The single-cell interface used for the comparison
88 with JumpProcesses.jl (Fig. 2e) allocates a workspace per simulation and is slower in serial execution than
89 JumpProcesses.jl on that one-gene model; the population interface reuses one workspace per thread.

90 Supplementary Table 3. Runtime of population simulations.

genes	cells	kernel	threads	seconds (200	
				steps)	cell-steps / s
10	100	DirectSSA	4	0.38	52,398
10	100	HybridSSATau	4	0.11	180,019
10	1000	DirectSSA	4	1.04	193,080
10	1000	HybridSSATau	4	0.58	346,004
10	5000	DirectSSA	4	4.78	209,388

genes	cells	kernel	threads	seconds (200 steps)	cell-steps / s
10	5000	HybridSSATau	4	2.57	388,683
50	1000	DirectSSA	4	9.72	20,581
50	1000	HybridSSATau	4	2.57	77,671
100	1000	DirectSSA	4	31.41	6,367
100	1000	HybridSSATau	4	6.37	31,373

Supplementary Note 8: Laboratory and clinical reference data

Supplementary Table 4. Reference values used for calibration and validation (transcribed from the cited articles; files in `paper/data/`).

Quantity	Value	Source
U2OS cells tracked at cisplatin addition (7 / 10 / 13 μ M)	232 / 240 / 296	¹ , Table 2
of which died within 72 h	30 / 66 / 176	same
of which divided within 72 h	100 / 75 / 67	same
of which survived without dividing	102 / 99 / 53	same
HCT116 cells at drug addition; death fraction	275; 0.64	¹ , Table 1
Plateau of the HCT116 kill curve	after ~100 h	¹ , Fig 1b
Lineage correlations of fate	present for sisters, first and second cousins; absent for third cousins	¹ , Fig 5a
Pre-resistant melanoma cells	1:50 to 1:500 per marker; EGFR-high cells give 7.9 ± 0.9 fold more resistant colonies	³
N15-0385 glioblastoma doubling time	50 h	⁴
Temozolomide elimination half-life	2.1 h	⁵
RTOG 0525 regimens	150-200 mg/m ² days 1-5 vs 75-100 mg/m ² days 1-21 of 28-day cycles; median OS 16.6 vs 14.9 months	⁶
SWOG S1320 regimens	continuous vs 3 weeks off / 5 weeks on after an 8-week lead-in; median PFS 9.0 vs 5.5 months	⁷

Supplementary Note 9: Migration from version 1

The v1 API (`Donne`, `ssa`, `tauleap`, `tauleapswitch`, `adaptive_tauleap`, `exponential_growth`, `NamedTuple` reactions) remains available through a deprecated compatibility layer that converts `NamedTuple` reactions to `ReactionModels` (reactions whose names contain `act`, `inhib`, `comb_a` or `comb_i` become Hill kinetics with `rate`

⁹⁸ = [k, n, K]; species whose names contain `on/off` become promoter groups) and maps the algorithms onto the
⁹⁹ new kernels. Output shapes are preserved, including the leading `:NULL` column. See `docs/src/migration.md`.

¹⁰⁰ Supplementary Note 10: Relation to previous work

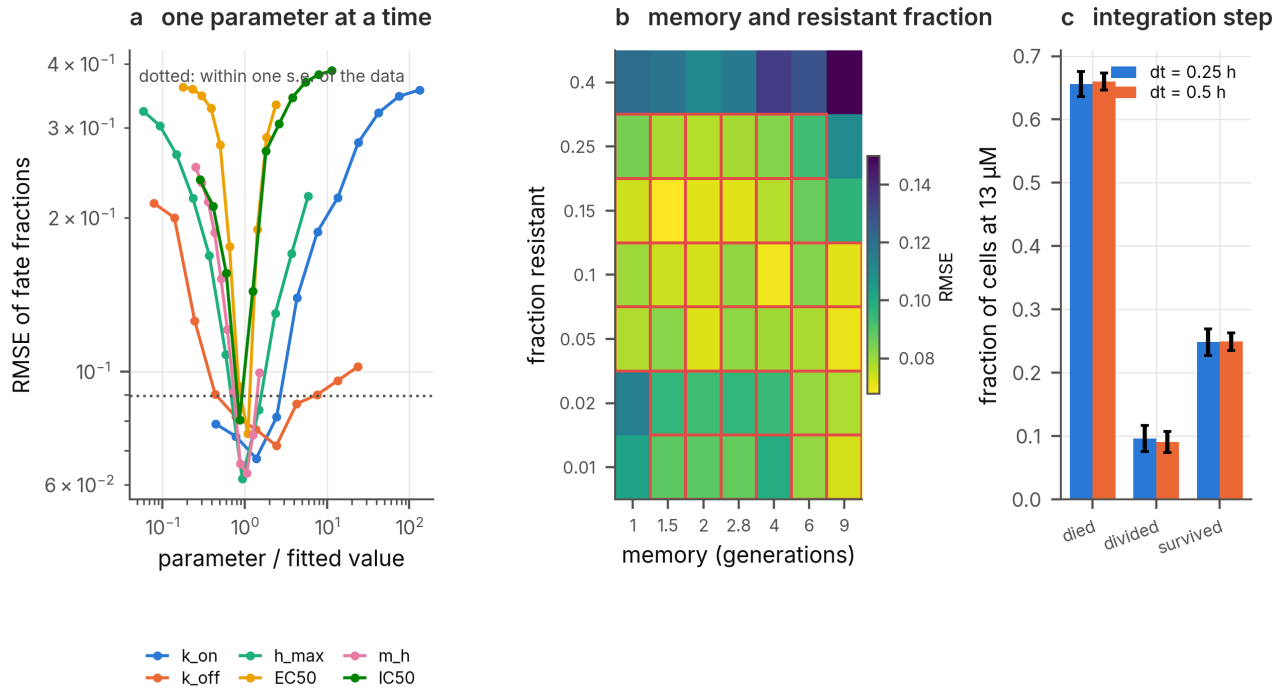
¹⁰¹ The ingredients of this framework have precedents, and several of its results confirm conclusions reached earlier
¹⁰² by other means. The table separates, claim by claim, what was already established from what this work adds.
¹⁰³ Prior results the framework reproduces are evidence that it is behaving correctly, not claims of novelty.

Result in this Article	Already established	What this work adds
Stochastic reaction kinetics inside growing, dividing cells (Fig. 3)	Simulators of stochastic expression with growth and division exist ^{8,9} , including version 1 of this framework ¹⁰	Gene replication, promoter inheritance, a drug layer, lineage statistics, observation models and inference in one forward model, so that a resistance phenotype is a heritable expression state with measurable kinetics rather than an assumed compartment
Simulation of division trees for lineage-tracing methods	TedSim couples expression to division history ¹¹ ; Cassiopeia simulates topologies, heritable fitness and CRISPR barcodes ¹²	Molecular content on the same tree: partitioning at division, promoter states, drug-induced death, and per-cell fates with the molecules that caused them
Heritable expression states decide which cells survive a drug (Fig. 4)	Measured directly in barcoded and time-lapse experiments ^{1,3,13,14}	A generative model in which memory is one promoter timescale, reproducing the reported signatures quantitatively and predicting which of them discriminate pre-existing from induced tolerance
Cell-cycle-aware inference of transcriptional kinetics is necessary (Fig. 3f, Fig. 6b)	Established from data and theory ^{15–17}	A simulator that generates the data such methods assume, and a quantification of the bias incurred by a division-blind fit
Benchmarks of network inference from single-cell data	Extensive benchmarks exist ^{10,18,19}	The growth and cell-cycle confound with known ground truth: how much accuracy is lost in a dividing population, and how much cell-cycle regression recovers (Fig. 5b)

Result in this Article	Already established	What this work adds
The benefit of treatment holidays depends on a cost of resistance and on turnover (Fig. 8a-e)	Compartment models of adaptive therapy ^{20,21}	The same dependence from single cells whose resistance is a heritable expression state; cost and degree of protection become cell properties that lineage experiments measure; partial protection reverses the ranking even without a cost
Perturbing the retention of a resistant state changes the outcome (Fig. 4f)	Phenotypic-switching theory ²² and memory-disrupting compounds ¹³	The timing requirement (disruption must continue during exposure) and a lineage-resolved readout, the number of surviving clones
An intermediate dose can be optimal against drug-induced tolerance (Fig. 4h)	Reported for induced persisters ²³	Reproduced; the accompanying benefit of release periods does not appear here, which the text attributes to the decay of the induced state in this model rather than to a disagreement about data
MGMT expression selects glioblastoma cells under temozolomide (Fig. 8f-h)	Phenotypic selection with stable inheritance ⁴	Pharmacokinetics, the RTOG 0525 regimens, consumption of MGMT by the drug, and the fractionation that minimises the final population at fixed cumulative dose
Dose-dense temozolomide does not improve survival	Clinical result ⁶	A mechanistic account of why, and identification of the rate at which the drug consumes MGMT as the measurement that separates the regimens

Supplementary Note 11: Identifiability and numerical robustness of the calibration

Six parameters were fitted to six measured fractions, so the fit cannot determine all six. Three checks bound what the calibration does and does not establish.



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109 **Supplementary Fig. 3 | What the fate fractions determine.** **a**, Root-mean-square error of the simulated
 110 fate fractions at the training concentrations when one parameter is moved away from its fitted value over its
 111 full range and the others are held at theirs, plotted against the ratio to the fitted value; the dotted line marks
 112 one binomial standard error of the measurement above the best fit. **b**, The same error over the memory of the
 113 resistant state and the fraction of cells in it, with the death parameters held at their fitted values; red outlines
 114 mark the combinations that lie within one standard error of the best. **c**, Simulated fates at 13 μM with the
 115 integration step halved (mean $\pm$ s.d. of three seeds).

116 The death parameters are the better determined: the maximal death rate stays within one standard error of the
 117 best fit over about a twofold range, and the EC_{50} , Hill coefficient and growth-arrest IC_{50} over narrower ranges
 118 still (Supplementary Fig. 3a). The two promoter switching rates are not determined: each can move by roughly
 119 sixfold with no penalty the data can detect. The two-dimensional slice shows why (Supplementary Fig. 3b). A
 120 short memory with many resistant cells and a long memory with few produce nearly the same fate fractions, so
 121 the error surface is flat along that direction, and a majority of the combinations tested, spanning one to nine
 122 generations of memory and 1% to 25% of cells resistant, fit within the noise of the measurement. The memory
 123 quoted in the main text is therefore set by the prior taken from the lineage correlations, not by the fate fractions,
 124 and the resistant fraction that accompanies it should not be read as a measurement.

125 Halving the integration step from 0.5 h to 0.25 h changes each simulated fate fraction by less than the seed-to-seed
 126 spread (Supplementary Fig. 3c), so the discretisation of the death hazard is not a source of error at this step size.

127 A profile in which the remaining parameters are re-optimised at each fixed memory would be preferable to a slice,
 128 but at seven seconds per objective evaluation the inner search could not be given enough budget to be reliable: it
 129 returned fits three times worse than the known solution at short memory, which reflects the optimiser rather than
 130 the data. The slice avoids that failure at the cost of conditioning on the fitted death parameters.

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